

When the vessel breaks

Pathophysiology of non-traumatic intracerebral haemorrhage. A review of Rossi and Cordonnier (2014), Chapter 5 of the *Oxford Textbook of Stroke and Cerebrovascular Disease*. Every figure is cited in APA style; full references on pages 4–5.

1 The vessel decides where it bleeds.

2 The first hours decide how big it gets.

3 The blood keeps injuring the brain for days.

01 The vessel

About 80–85% of ICH is **primary**: a small vessel weakened by chronic disease ruptures. About 15–20% is **secondary** to an identifiable lesion or condition (Macellari et al., 2014).

The two small-vessel diseases

(Fisher, 1971; Macellari et al., 2014; Rossi & Cordonnier, 2014)

	HYPERTENSIVE	AMYLOID (CAA)
Vessels	Deep perforators	Cortical, leptomeningeal
Wall	Lipohyalinosis; microaneurysms (debated)	β-amyloid replaces smooth muscle
Bleed site	Putamen, thalamus, pons, cerebellum	Lobar
Microbleeds	Deep	Strictly lobar
Patient	Long-standing hypertension	Usually aged 55+

Causes in one cohort (SMASH-U, n = 1013)

(Meretoja et al., 2012)

Hypertension	35%
Undetermined	21%
Amyloid angiopathy	20%
Anticoagulants (largest bleeds; 54% dead by 3 months)	14%
Structural lesion (AVM, cavernoma)	5%
Systemic disease (cirrhosis, low platelets)	5%

Microbleeds (tiny old leaks, black dots on blood-sensitive MRI) appear in 5% of healthy adults, 34% after ischaemic stroke and 60% of people with ICH (Cordonnier et al., 2007).

The patient in the talk. Jill Bolte Taylor, 37 and normotensive: an AVM ruptured on 10 December 1996. A textbook secondary ICH (Taylor, 2008a, 2008b).

Secondary causes: always ask

(Macellari et al., 2014)

-  **Vascular malformations:** AVM, cavernoma, dural fistula, aneurysm. Commonest in younger patients.
-  **Antithrombotic drugs:** make bleeds larger and keep them growing for longer.
-  **Tumours:** primary or metastatic; fragile new vessels.
-  **Cerebral venous thrombosis:** blocked drainage raises venous pressure.
-  **Haemorrhagic transformation** of an infarct.
-  **Others:** cocaine, amphetamines, vasculitis, RCVS, moyamoya, coagulopathy.

Red flags for a hidden lesion

(Macellari et al., 2014; Zhu et al., 1997)

- ✓ Aged 45 or under, or no history of hypertension
- ✓ Lobar bleed, or blood only in the ventricles
- ✓ CT clues: subarachnoid blood, calcification, abnormal vessels
- ✓ A bleed next to a venous sinus, or a dense sinus
- ✓ On anticoagulants: check clotting at once

65%

had a lesion on angiography: aged 45 or under, normotensive, lobar bleed

0%

had one: over 45, hypertensive, deep bleed (Zhu et al., 1997)

Modified Boston criteria for CAA

(Knudsen et al., 2001; Linn et al., 2010)

Definite	Full post-mortem shows severe CAA with a lobar bleed.
Probable	Age 55+; multiple lobar or cortical bleeds (or one plus cortical superficial siderosis); no other cause.
Possible	Age 55+; a single lobar bleed or siderosis; no other cause.

02 The first hours

01 Rupture. A diseased small artery gives way under arterial pressure.

03 Avalanche. The growing clot tears vessels at its edge (Fisher, 1971).

02 Haematoma. Blood splits the tissue planes.

04 Tamponade. Tissue pressure and clotting stop the bleed (Fisher, 1971).

38 in 100

grew by over a third within 20 h (26 within the first hour; Brott et al., 1997)

+5%

hazard of death for every 10% of growth (Davis et al., 2006)

⌚ Higher risk of growth

Early presentation, a large baseline bleed, anticoagulant use and a **spot sign**: contrast leaking into the clot on CT angiography (Brott et al., 1997; Demchuk et al., 2012).

○ Size in context

A golf ball holds about 40 mL. Volume of 60 mL or more with a GCS of 8 or less predicted 91% mortality at 30 days (Broderick et al., 1993). A clot that grows by a third is only about 10% wider on a scan, so growth is easy to miss by eye.

≈45%

extend into the ventricles: hydrocephalus risk and a worse outcome (Hanley, 2009)

40%

median case fatality at 1 month, unchanged 1980–2008 (van Asch et al., 2010)

03 The days after

WHEN	WHAT DRIVES THE OEDEMA (Xi et al., 2006)
Hours	Clot retraction squeezes serum into the tissue
Days 1–2	The coagulation cascade makes thrombin, which opens the blood–brain barrier
Day 3+	Red cells lyse; haemoglobin, haem and iron drive oxidative injury
Weeks	Macrophages clear the clot; a cavity and haemosiderin remain

Oedema grew by about 75% in the first 24 hours (Gebel et al., 2002), and it keeps building over the following days (Xi et al., 2006).

🔥 Three toxins, one result

⚠️ **Thrombin:** needed to stop the bleed, toxic at high levels.

⊗ **Iron:** released from lysed red cells; fuels free radicals.

🔥 **Inflammation:** microglia within hours, then neutrophils, macrophages, cytokines and MMP-9.

All three lead to a leaky barrier, more oedema and neuronal death. Most of this evidence comes from animal models (Xi et al., 2006); the first human iron-chelation trial was neutral (Selim et al., 2019).

Myth: the rim around the clot is starving tissue. PET at 5–22 h showed blood flow at 56% of the other side, but oxygen use at 48%. The tissue is resting, not starving (Zazulia et al., 2001). In ICH ADAPT, lowering systolic pressure below 150 mmHg did not reduce flow around the clot (Butcher et al., 2013).

04 At the bedside: every stage is a window

BEFORE AND AFTER

The vessel

Control blood pressure; it is the strongest tool against a first or second bleed. Review antithrombotics. Older adult with a lobar bleed: think CAA.

FIRST 24 HOURS

The first hours

Image fast, lower blood pressure early and reverse anticoagulants urgently. Keep neuro observations close; a falling GCS needs action now.

DAYS TO WEEKS

The days after

Expect fluctuation as oedema peaks. New drowsiness, headache, vomiting or a new deficit: stop and escalate. Reassess before every session and pace therapy.

These bedside points synthesise the evidence above. Full references (APA 7th edition): pages 4–5.

From mechanism to the bedside

What the pathophysiology predicts for assessment, timing and prognosis.

📍 Location predicts the impairment

PLIC: (Gupta et al., 2025; Murray et al., 2025; Puig et al., 2019)

LOCATION	WHAT YOU WILL SEE	PHYSIOTHERAPY FOCUS
Putamen, internal capsule	Opposite-side weakness ± sensory and field loss; aphasia (left) or neglect (right)	Check the PLIC on imaging; task-specific motor training
Thalamus	Opposite-side sensory loss, ataxia; later, central pain	Sensory and balance retraining; screen for new pain
Cerebellum	Same-side limb ataxia, truncal sway, vertigo, vomiting	Balance and gait. Drowsiness is an emergency (hydrocephalus)
Pons	Four-limb weakness, cranial nerve signs; often severe	Chest, positioning, seating; a communication plan
Lobar	By lobe: weakness, neglect, aphasia, field loss; seizures more common	Screen cognition and neglect; know the seizure plan

🕒 When to start: the trials

STUDY	FINDING	MAIN LIMIT
AVERT Trial Collaboration Group (2015) RCT · n = 2104	High dose within 24 h: fewer good outcomes (46% vs 50%)	ICH a minority; subgroup only
Bernhardt et al. (2016) AVERT dose analysis	More sessions a day: better (OR 1.13). More minutes: worse (OR 0.94)	Observational
Liu et al. (2014) RCT · n = 243	Rehab within 48 h: MBI +13, fewer deaths	HR CI 1.24–15.87; one country
Yen et al. (2020, 2021) RCT · n = 60	Out of bed at 24–72 h: better FIM-motor and FAC; BP stable	Small; mild–moderate only
Kan et al. (2026) 17 RCTs · n = 1396	Very early looked better than early	Low certainty

Practical reading. Start early once the team says the patient is stable. Keep sessions short and frequent. Avoid high-dose work in the first 24 hours.

⚠️ Before every session: stop and escalate if

- ✓ GCS down 2 or more, or new confusion or drowsiness
- ✓ New or worse headache, or vomiting
- ✓ New weakness, speech change, pupil change or a seizure
- ✓ BP outside the team's target for this patient
- ✓ Cerebellar bleed with any drowsiness: emergency

1

Notice the cues

2

Interpret with the timeline

3

Act: stop, escalate

4

Reflect and re-screen

Clinical reasoning cycle adapted from Levett-Jones et al. (2010). Day 3 onwards: oedema still building and iron released (Xi et al., 2006).

⚖️ Recovery: what to tell families

Matched on the **deficit**, bleeds recovered better (Paolucci et al., 2003; OR 2.48). Matched on **lesion size and site**, they did worse (Balk et al., 2026; OR 1.69 favouring infarct). Unmatched studies disagree (Kelly et al., 2003; Oosterveer et al., 2022; Salvadori et al., 2021). More of a bleed's deficit is pressure, which resolves, but the blood is toxic. A severe early motor deficit after a deep bleed is **not a fixed ceiling**; in severe putaminal bleeds most gains came in the first four months (Yoo & Chung, 2026).

🔍 Appraise any claim: five questions

- 1 Who was studied?** Survivors only? Mild cases only?
- 2 Compared with what?** Matched on deficit, or on lesion?
- 3 How big, how certain?** Read the interval, not just the p value.
- 4 Measured how?** Barthel and FIM have ceiling effects.
- 5 Does it fit my patient?** Setting, timing, severity.

📖 Glossary for junior colleagues

ICH	bleeding into brain tissue	Haematoma	the clot from the bleed	Oedema	swelling around the clot
CAA	amyloid in surface arteries	AVM	tangle of arteries and veins	Microbleed	tiny old bleed on MRI
Spot sign	contrast leak: still bleeding	IVH	blood in the ventricles	PLIC	where the motor fibres run
GCS	conscious level, 3 to 15	mRS	disability, 0 (none) to 6	FAC	walking ability, 0 to 5

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APA 7th edition. Every author is listed (up to 20; for 21 or more, the first 19, an ellipsis, then the final author).

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